

1 Discrete Choice Model

$$u_{ij} = x_j \tilde{\beta}_i - \alpha p_j + \xi_j + \varepsilon_{ij}$$

where

$$\tilde{\beta}_{ik} = \beta_k + \sigma_k \zeta_{ik} \text{ such that } \zeta_{ik} \sim \mathcal{N}(0, 1)$$

thus,

$$\underbrace{x_j \beta}_{\text{Avg. characteristic effects}} + \underbrace{\xi_j}_{\text{Unobserved characteristics}} - \underbrace{\alpha p_j}_{\text{Avg. price effect}} + \nu_{ij}$$

where

$$\nu_{ij} = \left[\sum_k x_{jk} \sigma_k \zeta_{ik} \right] + \varepsilon_{ij}$$

is a mean-zero, heteroskedastic error term that captures the effect of the random taste parameters

$$\underbrace{\delta_j \equiv x_j \beta - \alpha p_j + \xi_j}_{\text{Mean utility levels}}$$

Note that as a result of assuming an i.i.d. additive error term, we implicitly assume that any two pairs of goods with the same market share will have the same cross-price elasticities with any other third good.

However, modeling $\tilde{\beta}_i$ as a random coefficient avoids this assumption and produces reasonable substitution effects; i.e. consumers substitute towards goods with similar characteristics.

2 Discrete-Choice Market Share Function

Each consumer purchases one unit of the good that provide the highest utility

Conditonal on characteristics $(\vec{x}, \vec{\xi})$ and price $(\vec{p})$, consumer i will purchase good j if, and only if $\forall k \geq 0$ and $k \neq j$

$$U(x_j, \xi_j, p_j, \nu_i, \theta_d) > U(x_k, \xi_k, p_k, \nu_i, \theta_d)$$

Thus implicitly defines the set of unobservable taste parameters, ν_{ij} that result in the purchase of good j

Define the set of unobserved characteristics that lead to the purchase of good j as

$$A_j(\delta) = \{\vec{v}_i / \delta_k + \nu_{ij} > \delta_k + \nu_{ik} \text{ for all } k \neq j\}$$

Then, the marke share for firm j is the probabilitiy that nu_i falls into the region A_j

Given a distribution $F(\cdot, \vec{x}, \sigma)$ for $\vec{\nu}$ with density function $f(\cdot, x, \sigma)$, the market share is then

$$\mathcal{S}(\delta(\vec{x}, \vec{p}, \vec{\xi}), x, \theta) = \int_{A_j(\delta)} f(\vec{\nu}, \vec{x}, \sigma_\nu) d\nu$$

Thus we are integrating across the set of unobservable consumer tastes implicitly defined by A_j

3 Estimating from Mean Utility

Consider a demand equation $s_j = \mathcal{S}_j(\vec{x}, \vec{p}, \vec{\xi}, \vec{\theta})$, which relates predicted market shares to actual market shares.

The right side contains price and product level demand error implying prices are endogenous as they are correlated with the unobserved characteristics.

The usual IV doesn't work since the unobserved characteristics enter in a non-linear way.

Starting at the simplest case, suppose that the distribution of consumer unobservables was known, so market shares depend only on mean utility levels;

$$s_j = \mathcal{S}_j(\vec{\delta})$$

So, at the true values, $\vec{s}$ and $\vec{\delta}$ this equation holds exactly.

Note the importance of the fact that the mean utility levels δ_j contain the aggregate error ξ_j , therefore, conditional on the true value of $\vec{\delta}$, the model should fit the data exactly

This exact fit can be exploited. If the equation $\vec{s} = \mathcal{S}(\vec{\delta})$, can be inverted, it yields $\vec{\delta} = \mathcal{S}^{-1}(\vec{s})$, so the observed market shares uniquely determined the mean utility for each good.

See the appendix, which shows that, under weak regularity conditions, there can exist one, and only one mean utility level for each value of $\vec{s}$.

For any density $f(\cdot, x)$, we can therefore calculate $\vec{\delta}$ from observations of the market shares alone.

The unique calculated value $\delta(s)$ can be used in an estimation procedure. When the density of ν is known exactly, so that the market share function depends on no unknown parameters, other than the vector $\vec{\delta}$, then the calculated mean utility levels can be treated as a known nonlinear transformation of market shares, $\vec{s}$.

For the true values of (β, α) ,

$$\delta_j(s) = x_j\beta + \alpha p_j + \xi_j$$

then we can use an IV regression of $\delta_j(s)$ on (x_j, p_j) to estimate (β, α) , treating ξ_j as an unobserved error term.

The correlation of p_j and ξ_j suggest using instruments for prices. Cost variables that are excluded from x_j , i.e. input prices that vary across firms are appropriate instruments.

With differentiated products, the characteristics of other firms $(x_k, k \neq j)$ are also valid instruments. Since they are not entered into u_{ij} but are correlated with prices via the markup in the FOC's

4 Market Size

Either explicitly observed or left as a parameter to estimate.

Observed output quantity of the firm is then given by

$$q_j = M\mathcal{S}(\vec{x}, \vec{\xi}, \vec{p}, \theta_d)$$

5 The Outside Good

As well as the $j = (1, 2, \dots, N)$ firms we assume that there exists an outside firm $j = 0$ whose price does not depend on the prices of the inside goods.

Thus, a general price increase will decrease aggregate demand, unlike previous works.

But with $N + 1$ firms, $(q_1, \dots, q_N)$ is not enough to determine market share for the $N + 1$ firms.

We assume we observe or can estimate M , then the market shares are given by,

$$s_j = q_j/M$$

6 The Supply Side

Assume that the N firms are price setters (this can be adapted to alternate models).

Total cost for firm j is given by the function $C_j(q_j, w_j, \omega_j, \vec{\gamma})$ and marginal cost are $c_j(q_j, w_j, \omega_j, \vec{\gamma})$ where $\vec{\gamma}$ is a vector of unknown parameters. Then profits for firm j is;

$$\pi_j = (\vec{p}, \vec{z}, \vec{\xi}, \omega_j, \theta) = p_j M \mathcal{S}(\vec{x}, \vec{\xi}, \vec{p}, \theta_d) - C_j(q_j, w_j, \omega_j, \vec{\gamma})$$

where $\theta = (\theta_d, \vec{\gamma})$

Assuming the existence of a pure strategy, interior equilibrium, the price vector satisfies the usual FOC:

$$[p_j - c_j] \left[\frac{\partial \mathcal{S}_j}{\partial p_j} \right] + \mathcal{S}_j = 0$$

or equivalently

$$p_j = c_j + \mathcal{S}_j / \left| \frac{\partial \mathcal{S}_j}{\partial p_j} \right|$$

Then if the N such FOC's define a unique equilibrium for all possible values of the observed and unobserved data and all possible values of the parameters, then the FOC's implicitly define a reduced-form price $\mathcal{P}_j(\vec{z}, \vec{\xi}, \vec{\omega}, \theta)$ as a function of exogenous data and the parameters.

Together with the demand function we get a reduced-form expression for equilibrium quantities:

$$q_j(\vec{z}, \vec{\xi}, \vec{\omega}, \theta) = M \mathcal{S}_j(\vec{x}, \vec{\xi}, \mathcal{P}_j(\vec{z}, \vec{\xi}, \vec{\omega}, \theta), \theta_d)$$

7 Estimating Density Parameters

We have s_0 for assumed the vector of consumer characteristic is known.

Now we proceed assuming that the density of ν_{ij} (the consumer characteristic), $f(\cdot, x, \sigma)$ depends on a vector of unknown parameters σ , which are to be estimated.

We are rarely concerned with σ directly but are rather concerned with distributional assumption on σ that will yield unreasonable or inaccurate estimates of economically interesting variables.

Upon parameterizing the distribution of consumer characteristics, σ , the market share function and subsequently the implied mean utility levels will also now vary with σ .

The mean utility levels now implicitly defined by $s = \mathcal{S}(\delta, \sigma)$ yields;

$$\delta_j(\vec{s}, \sigma) = x_j\beta + \alpha p_j + \xi_j$$

Now with the entry of σ into our estimation we can continue to use IV techniques to estimate (α, β, σ) .

Though, since σ may enter in a non-linear fashion we should use the generalized method of moments. Further, since we now must estimate an new variable we require another instrument.

8 Example - Nested Logit

Nested logit allows for the assumption that consumer tastes have an extreme value distribution but, allows for consumer tastes to be correlated across product, j , allowing for more reasonable substitution patterns across productions.

Procedure:

1. Group products into $G + 1$ exhaustive and mutually exclusive groups, $g = 0, 1, \dots, G$. Denote the set of products in group g by $\mathcal{J}_g$. The outside good, $j = 0$ is assumed to be the only member of group 0.
2. For $j \in \mathcal{J}_g$ assume that the utility of consumer i is

$$u_{ij} = \delta_j + \zeta_{ig} + (1 - \sigma)\varepsilon_{ij}$$

where, once again,

$$\delta = x_j\beta - \alpha p_j + \xi_j$$

and ε is i.i.d. extreme value. For consumer i , ζ is common to all products in group g and have a distribution function dependent on $\sigma \in [0, 1)$. The distribution of ζ is the unique distribution such that if ε is an extreme value r.v., then $[\zeta + (1 - \sigma)\varepsilon]$ is also an extreme value r.v.

We can interpret our utility function

$$u_{ij} = \delta_j + \zeta + (1 - \sigma)\varepsilon_{ij}$$

as a random coefficient model involving random coefficients ζ_{ig} on group-specific dummy variables as $d_{jg} = \mathbb{1}\{j \in \mathcal{J}_g\}$.

This our model can be written as;

$$u_{ij} = \delta_j + \sum_g [d_{jg}\zeta_{ig}] + (1 - \sigma)\varepsilon_{ij}$$

Unlike a more general random coefficient models, the nested logit allows correlation patterns to depend only on groupings of products that are determined prior to estimation and not on the values of continuous variables.

Using known formula for market share (page 253) yields the following linear instrumental variable regression

$$\ln(s_j) - \ln(s_0) = x_j\beta - \alpha p_j + \sigma \ln(\bar{s}_{j/g}) + \xi_j$$

where $\bar{s}_{j/g}$ is within group market share.

Since $\bar{s}_{j/g}$ is clearly endogenous, we need an instrument to instrument for $\bar{s}_{j/g}$, such as characteristics of other firms in the group.